

D.2 Electric and magnetic fields

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **26 marks**

$k = 8.99 \times 10^9 \text{ N m}^2\text{C}^{-2}$. Show your working.

Question 1 — The balloon war

(8 marks)

After vigorous rubbing on two jerseys, balloon A carries $+4.0 \mu\text{C}$ and balloon B carries $+1.0 \mu\text{C}$. Their centres are 0.30 m apart.

- (a) State Coulomb's law. [2]
- (b) Calculate the electrostatic force between the balloons. [3]
- (c) Balloon A carries four times the charge of B. Compare the force A exerts on B with the force B exerts on A, and justify your comparison. [3]

Question 2 — The tug-of-war in the middle

(6 marks)

Charges $+q$ and $-q$ are fixed a distance d apart.

- (a) A student announces that the electric field at the midpoint is zero, “because the charges are opposite and cancel”. By considering the *direction* of each charge's field at the midpoint, explain why the student is wrong. [3]
- (b) State the direction of the resultant field at the midpoint. [1]
- (c) Now both charges are made $+q$. State the field at the midpoint in this case, and reconcile the two answers. [2]

Question 3 — The wires in the wall

(6 marks)

Two long, straight, parallel wires behind a wall carry currents in *opposite* directions, and repel with force F per unit length.

- (a) Explain, using the magnetic field of one wire and the force on the other, why antiparallel currents repel. [3]
- (b) During renovation both currents are reversed. State the new force per unit length and its character (attractive or repulsive), with a reason. [2]
- (c) State what *would* make the wires attract. [1]

Question 4 — The rulebook of field lines

(6 marks)

- (a) State two rules obeyed by electric field lines, and give the reason field lines can never cross. [3]
- (b) Two parallel plates 2.0 mm apart have a potential difference of 600 V across them. Calculate the electric field strength between them, and state its direction relative to the plates. [3]

Answers

Question 1 — The balloon war

- (a) The electrostatic force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them, acting along the line joining them.
- (b) $F = \frac{kq_1q_2}{r^2} = \frac{(8.99 \times 10^9)(4.0 \times 10^{-6})(1.0 \times 10^{-6})}{(0.30)^2} = 0.40 \text{ N}$, repulsive since both charges are positive.
- (c) They are *equal* in magnitude and opposite in direction. Coulomb's law contains the *product* q_1q_2 , which is the same whichever balloon you consider, and Newton's third law requires it. The unequal charges set the size of the single shared force; they do not give each balloon its own.

Question 2 — The tug-of-war in the middle

- (a) Field direction is defined by the force on a *positive* test charge. At the midpoint, the field of $+q$ points *away* from $+q$, that is, towards $-q$. The field of $-q$ points *towards* $-q$, which is the *same* direction. The two therefore reinforce rather than cancel. Each has magnitude $\frac{kq}{(d/2)^2} = \frac{4kq}{d^2}$, so the resultant is $\frac{8kq}{d^2}$, its maximum, not zero. The student has confused opposite *charges* with opposite *field directions*.
- (b) Along the line joining the charges, from $+q$ towards $-q$.
- (c) Now the field is *zero*. Two like charges produce fields at the midpoint of equal magnitude pointing in *opposite* directions, so they cancel. The student's slogan was right about the wrong configuration: cancellation happens for *like* charges, reinforcement for opposite ones.

Question 3 — The wires in the wall

- (a) Each wire sits in the magnetic field of the other. Wire 1 produces circular field lines around itself, and at the position of wire 2 that field is perpendicular to wire 2's current. The force $F = BIL$ on wire 2 is therefore perpendicular to both, and with the currents *antiparallel* it points away from wire 1. The same argument applied to wire 1 gives an outward force there too, so the pair repels.
- (b) Still F , and still repulsive. Reversing *both* currents leaves their *relative* orientation antiparallel, and the force depends on that relative orientation, not on the absolute direction of either current.
- (c) Reversing only *one* of the currents, which would make them parallel. Parallel currents attract.

Question 4 — The rulebook of field lines

- (a) Any two of: lines begin on positive charge and end on negative charge; their spacing indicates the field strength, closer together meaning stronger; the tangent to a line gives the field direction at that point. They can never cross, because a crossing point would assign two different field directions to one point, and the field there has a single well-defined direction.
- (b) $E = \frac{V}{d} = \frac{600}{2.0 \times 10^{-3}} = 3.0 \times 10^5 \text{ V m}^{-1}$, directed perpendicular to the plates, from the positive plate towards the negative.